

Further Progress on the Alaoglu–Erdős Two-Base Exponent Conjecture

Multiplicative rigidity, transcendental product exponents,
primitive-power statistics, ambient root saturation,
and a certified search through 2^{26}

Research report prepared for Vladimir Reshetnikov

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Abstract

Let

$$\mathcal{S} = \{x \in \mathbb{R} : 2^x \in \mathbb{Z}, 3^x \in \mathbb{Z}\}.$$

The Alaoglu–Erdős conjecture asserts that $\mathcal{S} = \mathbb{N}_0$. This report starts from the Lean development and the combined report in the `ProveIt` repository at commit `5d3982ceb12644fdd3499569d9db5f62f9a31dc6`. No unconditional proof of the conjecture is obtained. Its final obstruction remains a special case of the unresolved four-exponentials problem. The investigation nevertheless produces several new exact consequences and equivalent formulations.

Conditional on a counterexample, the repository proves that there is a transcendental least noninteger solution β for which every solution has unique coordinates

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}.$$

We prove that this monoid is rigid under every rational function with algebraic coefficients: if $R \in \overline{\mathbb{Q}}(T)$ is regular at β , then $R(\beta) \in \mathcal{S}$ exactly when $R(T) = n + kT$ for some $n, k \in \mathbb{N}_0$. In particular, for arbitrary $x, y \in \mathcal{S}$,

$$xy \in \mathcal{S} \iff x \in \mathbb{N}_0 \text{ or } y \in \mathbb{N}_0.$$

Thus the conjecture is equivalent to multiplicative closure of $\mathcal{S}$, and even to closure under squaring alone.

A second argument combines the repository’s three-smooth-base theorem with the classical six-exponentials theorem. If $x, y \in \mathcal{S}$ are both nonintegral, then 6^{xy} is transcendental; each of 2^{xy} and 3^{xy} is either an integer or transcendental, and at least one is transcendental. Hence failure of multiplicative closure is not a one-unit integrality miss: it is witnessed by genuine transcendence.

The exact coordinates also classify every common perfect-power and affine-descent pattern. If $x = n + k\beta$, the two outputs are simultaneously perfect r -th powers exactly when $r \mid n$ and $r \mid k$; their maximal common power degree is $\gcd(n, k)$. Consequently the proportion of simultaneously power-primitive output pairs of height below T tends to $1/\zeta(2) = 6/\pi^2$, and the exact degree g occurs with limiting probability $6/(\pi^2 g^2)$. We further identify the candidate divisibility lattice, prove unconditional closure under $\gcd$ and lcm , enumerate candidate divisors, and compute the ambient root saturation in terms of the primitive odd core and the rational-power index.

Finally, a five-thread MPFR 4.2.2 verifier using 192-bit directed rounding checks every non-power of two below 2^{26} , a total of 67,108,837 inputs. No candidate occurs. At the explicitly stated software trust level, every hypothetical noninteger solution therefore satisfies $x > 26$. A variable-order optimization of the formalized finite-difference obstruction explains quantitatively why that method, by itself, still falls short. The report ends with a prioritized formalization and research program.

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1 Executive summary

1.1 Bottom line

The conjecture remains unproved. This conclusion is not a failure of effort but a precise mathematical diagnosis. The repository already reduces any counterexample to a canonical primitive radical and a free rank-two solution monoid. Every elementary counting, progression, descent, and equidistribution constraint currently known is internally consistent with that monoid. To finish the proof one needs a new input at the four-exponentials boundary, or an argument that uses the special *integrality and valuation structure* more strongly than general transcendence theory presently does.

The strongest additions of this report are:

1. an exact rational-function intersection theorem for the canonical generator;
2. unconditional multiplication, quotient, and power criteria inside $\mathcal{S}$;
3. algebraicity and transcendence reformulations based on six exponentials;
4. a complete common-perfect-power and matched-affine-decomposition classification;
5. primitive-output asymptotics and exact limiting degree statistics;
6. a candidate gcd/lcm lattice and exact ambient root saturation;
7. the optimized variable-order scale $n/(\theta \log n)$ for the higher-difference method, together with a sharp quantitative explanation of its gap;
8. a directed-rounding finite exclusion through 2^{26} .

1.2 Trust labels

The report keeps four proof levels separate.

[kernel-verified Lean]	Accepted by the Lean kernel in the inspected repository snapshot.
[paper proof in this report]	A conventional proof is supplied here; it has not yet been checked by Lean.
[classical theorem used]	The proof invokes a named published theorem, principally six exponentials.
[directed- rounding computation]	A finite statement checked by directed-rounding software; this is not a kernel proof.

The finite search is useful evidence and a stronger explicit bound, but it does not change the logical status of the infinite conjecture. The kernel-verified bound remains $x \geq 12$ for a noninteger solution; the new $x > 26$ bound carries the disclosed software trust boundary.

2 The problem and the audited repository snapshot

2.1 Original and candidate forms

Conjecture 2.1 (Alaoglu–Erdős, two-base integer form). *For every real number x ,*

$$2^x \in \mathbb{Z} \quad \text{and} \quad 3^x \in \mathbb{Z} \quad \implies \quad x \in \mathbb{Z}.$$

The hypotheses force $x \geq 0$, so the intended conclusion is $x \in \mathbb{N}_0$. Put

$$\theta := \log_2 3 = \frac{\log 3}{\log 2} = 1.58496250072115618145 \dots$$

and define the natural-candidate set

$$\mathcal{C} := \{m \in \mathbb{N}_{>0} : m^\theta \in \mathbb{N}\}.$$

The maps $x \mapsto 2^x$ and $m \mapsto \log_2 m$ give inverse bijections between $\mathcal{S}$ and $\mathcal{C}$, because

$$(2^x)^\theta = 3^x.$$

Integral solutions correspond exactly to powers of two. Thus

$$\text{Conjecture 2.1} \iff \mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.$$

A positive natural number will be called *three-smooth* when all its prime factors belong to $\{2, 3\}$, equivalently when it has the form $2^u 3^v$. The set of such numbers is denoted $\mathcal{H}_{2,3}$.

2.2 Repository snapshot

The inspected branch is `main` at commit

5d3982ceb12644fdd3499569d9db5f62f9a31dc6,

whose commit time is 21 August 2026 at 02:43:24 UTC, still 20 August in Pacific time. The starting document is

Analysis/ExponentialIdentities/Article/
alaoglu-erdos-combined-report/
alaoglu-erdos-combined-report.tex.

Its accompanying formal development lies under

Analysis/ExponentialIdentities/Lean/
ExponentialIdentities/.

The combined report is already a substantial synthesis: it records the three-base theorem, Gelfond–Schneider consequences, exact finite certificates through 4096, rank-two and common-odd-core structure, the canonical primitive generator, exact counting and Fourier block averages, output normalization, localized-radical equivalences, exact radical degree, and arbitrary-order finite-difference obstructions. The present report treats those facts as the established frontier and asks what further consequences can be forced from them.

2.3 Why this is a four-exponentials problem

If x were a noninteger solution, then $1, x$ would be linearly independent over $\mathbb{Q}$, and $\log 2, \log 3$ would also be linearly independent over $\mathbb{Q}$. Yet all four exponentials

$$e^{\log 2} = 2, \quad e^{\log 3} = 3, \quad e^{x \log 2} = 2^x, \quad e^{x \log 3} = 3^x$$

would be algebraic. The four-exponentials conjecture would rule this out immediately. The six-exponentials theorem succeeds once a third independent logarithm or exponent is available, but the 2×2 case remains open [8, 9, 11, 12].

This observation is a methodological firewall. A proposed argument that uses only the algebraicity of the four displayed values is, in effect, attempting a new proof of a four-exponentials case. A more plausible route must exploit features absent from the general conjecture: positivity, integrality, prime valuations, exact denominator support, or the canonical least-output normalization.

3 The formalized frontier used in this report

This section states the repository results that enter the new arguments. It is deliberately selective: the combined report remains the detailed inventory, while the goal here is to make every dependency of the new theorems visible.

3.1 Elementary restrictions and transcendence

The development proves the following.

- [\[kernel-verified Lean\]](#) If $2^x \in \mathbb{Z}$, then $x \geq 0$.
- [\[kernel-verified Lean\]](#) If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Hence every noninteger solution is irrational.
- [\[kernel-verified Lean\]](#) If $2^x \in \mathbb{Z}$, then x is either an integer or transcendental over $\mathbb{Q}$. In particular every noninteger two-base solution is transcendental. The headline theorem is `transcendental_of_not_integer_of_two_three_rpow_integer` in `Transcendence.lean`.
- [\[kernel-verified Lean\]](#) The constant $\theta = \log_2 3$ is transcendental. Indeed $2^\theta = 3$, while $1 < \theta < 2$, so the same Gelfond–Schneider dichotomy applies.

The full Gelfond–Schneider theorem is formalized in the repository, following the Mathlib formalization of Karatarakis. The new arguments below use only the two consequences just listed, not the internal auxiliary-function proof.

3.2 Kernel-verified finite region

[\[kernel-verified Lean\]](#) The finite-region proof is distributed across three modules:

`FiniteCheck.lean`, `FiniteCheck256.lean`, `FiniteCheck4096.lean`.

Together they establish

$$m \in \mathcal{C}, \quad m < 4096 = 2^{12} \quad \implies \quad m \text{ is a power of } 2.$$

Equivalently,

$$x \in \mathcal{S} \setminus \mathbb{N}_0 \quad \implies \quad x \geq 12.$$

The 4096 theorem is replayed by the kernel from a table of exact integer comparisons using rational enclosures for θ . It contains no `sorry` and does not use `native_decide`. This is a stronger trust model than an interval scan.

The combined report also described an independent software-level scan through 2^{20} . [Section 11](#) replaces it by a directed-rounding MPFR verifier and extends the range sixty-fourfold, through 2^{26} .

3.3 Rank two, the primitive odd core, and the exact solution monoid

The three-base determinant theorem implies that prime-valuation vectors of candidates have rational rank at most two. Since 2 is already a candidate, every odd part lies on one rational ray. The repository normalizes that ray to a unique primitive odd core.

The decisive theorem for the present report is

`exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture`

from `PrimitiveGenerator.lean`.

Theorem 3.1 (Canonical primitive generator, repository theorem). *[kernel-verified Lean] Assume that the conjecture fails. Then there exist natural numbers w, d, a, A and a real number β such that:*

- (i) $w > 1$ is odd and its positive odd-prime valuations have gcd one;
- (ii) $d > 0$, $A > 0$, and $a = 0$ or $3 \nmid A$;
- (iii) with $E = 3^{\log_2 w} = w^\theta$,

$$E^d = \frac{A}{3^a}, \quad E^j \in \mathbb{Q} \iff d \mid j \quad (j \in \mathbb{Z});$$

(iv)

$$\beta = a + d \log_2 w, \quad M := 2^\beta = 2^a w^d \in \mathbb{N}, \quad 3^\beta = A \in \mathbb{N};$$

- (v) β is the unique least noninteger solution and is irrational;
- (vi) every solution has a unique coordinate pair

$$x = n + k\beta, \quad n, k \in \mathbb{N}_0;$$

(vii) every candidate has a unique coordinate pair

$$m = 2^n M^k, \quad n, k \in \mathbb{N}_0.$$

Combining the theorem with the formalized Gelfond–Schneider consequence gives:

Corollary 3.2. *[kernel-verified Lean] Under failure, β is transcendental over $\mathbb{Q}$, and hence also over $\overline{\mathbb{Q}}$.*

The last implication uses transitivity of algebraicity: a number algebraic over the algebraic closure $\overline{\mathbb{Q}}$ would be algebraic over $\mathbb{Q}$.

The theorem says much more than “rank at most two.” It identifies the entire solution set with the free additive monoid on $1, \beta$, the candidate set with the free multiplicative monoid on $2, M$, and the output-pair monoid with the free monoid on $(2, 3)$ and (M, A) . There are no omitted lattice points or Frobenius gaps.

3.4 Least-output and simultaneous normalization

[kernel-verified Lean] The least solution is affinely primitive: if

$$M = 2^r U^q, \quad A = 3^r V^q$$

with $q > 0$, then $r = 0$ and $q = 1$. In particular M, A are not both perfect q -th powers for any $q \geq 2$, and not both divisible by their distinguished bases.

The stronger simultaneous normalization in `OutputNormalization.lean` writes

$$M = 2^a w^d, \quad A = 3^b v^e,$$

where w is odd and power-primitive, v is not divisible by 3 and is power-primitive, $d, e > 0$, and

$$a = 0 \quad \text{or} \quad b = 0, \quad \gcd(\gcd(d, e), |b - a|) = 1.$$

The depths a, b and outer degrees d, e are intrinsic valuation invariants of the least output pair, not artifacts of a chosen factorization.

3.5 Exact radical endpoint

[kernel-verified Lean] The conjecture is equivalent to the localized-radical exclusion

$$u^\theta \notin \mathbb{Z}[1/3] \quad \text{for every odd } u > 1.$$

It suffices to consider power-primitive odd u . For the canonical core w , the least index d above is also the exact degree of $E = w^\theta$ over $\mathbb{Q}$: the binomial

$$X^d - \frac{A}{3^a}$$

is irreducible and is the minimal polynomial of E . Thus a counterexample creates a very special positive radical whose only rational-power denominator is supported at the prime 3.

Taking logarithms gives the bilinear relation

$$d \log w \log 3 = (\log A - a \log 3) \log 2. \quad (1)$$

Baker’s theory controls linear forms in logarithms with algebraic coefficients; it does not exclude this product-of-logarithms relation. The four-exponentials conjecture does.

3.6 Three-smooth classification of auxiliary bases

The formalized three-base determinant argument has the following exact consequence.

Theorem 3.3 (Three-smooth-base theorem, repository theorem). [kernel-verified Lean] *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$, and let $B \in \mathbb{N}_{>0}$. Then*

$$B^x \in \mathbb{Z} \iff B = 2^u 3^v \text{ for some } u, v \in \mathbb{N}_0.$$

The reverse implication is elementary. The forward implication is a specialized six-exponentials determinant proof: a prime divisor of B outside $\{2, 3\}$ supplies a third multiplicatively independent base and forces x to be rational, hence integral.

3.7 Counting and local progression obstructions

[kernel-verified Lean] Exact coordinates give exact unit-block and dyadic-block counts. In particular, if failure occurs then

$$N_\beta(T) := \#\{(n, k) \in \mathbb{N}_0^2 \setminus \{(0, 0)\} : n + k\beta < T\} = \frac{T^2}{2\beta} + O_\beta(T). \quad (2)$$

The repository proves the corresponding sharp limit, an equivalence between the conjecture and subquadratic candidate growth below 2^T , and convergence of exact block averages for every finite trigonometric polynomial. The continuous-test and interval-indicator extension is a paper-level Weyl consequence in the combined report.

[kernel-verified Lean] For every $r \geq 2$, set

$$c_r(\theta) := |\theta(\theta - 1) \cdots (\theta - r + 1)|.$$

Candidates cannot occupy all points

$$n, n + h, \dots, n + rh$$

when

$$c_r(\theta)h^r < n^{r-\theta}. \quad (3)$$

This is proved by an iterated mean-value theorem for forward differences: the r -th difference is a nonzero integer but has absolute value below one. We optimize the order r in [section 10](#).

4 Attempts at a full proof and their exact failure points

The following routes were pushed far enough to identify a precise obstruction. Recording the obstruction is important: it prevents a reformulation already compatible with the exact monoid from being mistaken for a proof.

4.1 Shrinking the three-base determinant to two bases

The formalized determinant method has three independent logarithmic directions against two exponent directions. Its arithmetic lower bound and analytic interpolation upper bound then have enough dimensional slack to force a contradiction. With only $\log 2, \log 3$, the balance lands at the classical four-exponentials threshold. Repackaging the same grid, changing a few exponents, or using a larger square matrix does not create the missing dimension. A successful two-base determinant must use additional arithmetic information, such as simultaneous non-Archimedean divisibility or the exact primitive normalization.

4.2 The radical endpoint

The reduction to an odd power-primitive w with $w^\theta \in \overline{\mathbb{Q}}$, indeed with an exact binomial minimal polynomial, is extremely sharp. It does not itself contradict a known theorem. Even the first-looking special value

$$3^{\log_2 3} = e^{(\log 3)^2 / \log 2}$$

is not presently known to be irrational. Gelfond–Schneider requires an algebraic irrational exponent; here $\log_2 3$ is transcendental. Linear forms in logarithms do not control the quadratic logarithmic expression.

4.3 Equidistribution against exponentially shrinking targets

A counterexample generates the irrational rotation orbit $\{k\beta\}$, and the exact block description makes that orbit equidistributed. The integrality gap at height k , however, is exponentially small, of scale roughly M^{-k} . Ordinary discrepancy and Dirichlet approximation control fixed or polynomially shrinking intervals, not this exponential scale. Equidistributed sequences with bounded partial quotients can avoid such targets indefinitely. A contradiction would require Diophantine information special to a β satisfying $2^\beta, 3^\beta \in \mathbb{N}$, not generic rotation theory.

4.4 Counting and additive combinatorics

Under failure the candidate set has only $O(\log X)$ points in $[X, 2X)$. This is far below the density regimes of Roth, Szemerédi, or standard quantitative progression results. General S -unit theory can show finiteness of many projective additive patterns, but the required conclusion is nonexistence. The optimized finite-difference analysis below makes the mismatch quantitative.

4.5 Descent and perfect powers

The least output pair has no common depth, common perfect-power degree, or matched affine decomposition. Yet the exact monoid predicts infinitely many later pairs with such features. Descent simply divides their coordinate pair by a common factor and returns to another allowed point. To close the problem, descent must produce a point below the canonical least generator; the exact coordinate criteria in [section 7](#) show exactly when that can and cannot happen.

4.6 A proof-audit criterion

Any claimed proof must add an ingredient not already implied by $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$. The following facts are all compatible with a counterexample and cannot alone settle the conjecture:

- zero density or logarithmic-square counting;
- equidistribution of fractional parts;
- uniqueness of the primitive odd core;
- transcendence of every noninteger solution;
- arbitrary but finite computational exclusion;
- fixed-order or variable-order local progression obstructions without a mechanism forcing the required progression.

5 Rational-function rigidity and nonlinear closure

The exact generator theorem has a consequence not isolated in the starting report: a hypothetical solution monoid is rigid against every nonlinear rational operation over the algebraic numbers.

5.1 The rational-function intersection theorem

Theorem 5.1 (Rational-function rigidity at the primitive generator). *[paper proof in this report] Assume failure, and let β be the canonical generator from theorem 3.1. Let $R(T) \in \overline{\mathbb{Q}}(T)$ have no pole at β . Then*

$$R(\beta) \in \mathcal{S} \iff R(T) = n + kT \text{ in } \overline{\mathbb{Q}}(T) \text{ for some } n, k \in \mathbb{N}_0.$$

Proof. The reverse implication follows from additive closure: $R(\beta) = n + k\beta \in \mathcal{S}$.

For the forward implication, write $R = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[T]$ and $Q(\beta) \neq 0$. If $R(\beta) \in \mathcal{S}$, exact coordinates give $R(\beta) = n + k\beta$ for some $n, k \in \mathbb{N}_0$. Therefore

$$P(\beta) - (n + k\beta)Q(\beta) = 0.$$

By theorem 3.2, evaluation at β is injective on $\overline{\mathbb{Q}}[T]$. Hence $P(T) - (n + kT)Q(T)$ is the zero polynomial, proving the claim. $\square$

This theorem permits poles and cancellation and allows the full algebraic closure as coefficient field. It can be read as the exact intersection

$$\mathcal{S} \cap \overline{\mathbb{Q}}(\beta) = \mathbb{N}_0 + \mathbb{N}_0\beta,$$

with the additional information that every rational expression realizing an element of the intersection is identically the corresponding affine expression.

Corollary 5.2 (Rational self-maps). *[paper proof in this report] Under failure, the rational functions over $\overline{\mathbb{Q}}$ that are regular on $\mathcal{S}$ and map $\mathcal{S}$ into itself are exactly*

$$T \mapsto n + kT, \quad n, k \in \mathbb{N}_0.$$

Proof. Such a function sends β to $\mathcal{S}$, so theorem 5.1 identifies it. Conversely, natural translations and natural dilations preserve $\mathbb{N}_0 + \mathbb{N}_0\beta$. $\square$

More generally, if $x_i = n_i + k_i\beta \in \mathcal{S}$ and $F \in \overline{\mathbb{Q}}(X_1, \dots, X_r)$ is regular at the tuple, then

$$F(x_1, \dots, x_r) \in \mathcal{S}$$

exactly when the one-variable specialization

$$F(n_1 + k_1T, \dots, n_r + k_rT)$$

is identically $p + qT$ with $p, q \in \mathbb{N}_0$. Thus every rational operation on finitely many solutions is reduced to an elementary identity in one indeterminate.

5.2 Exact multiplication criterion

Theorem 5.3 (Products of two solutions). *[paper proof in this report] For all $x, y \in \mathcal{S}$, unconditionally,*

$$xy \in \mathcal{S} \iff x \in \mathbb{N}_0 \text{ or } y \in \mathbb{N}_0.$$

Proof. If one factor, say x , is a nonnegative integer, then xy is a natural-number multiple of a solution, so $xy \in \mathcal{S}$.

For the converse, there is nothing to prove if the conjecture is true. Under failure write uniquely

$$x = n + k\beta, \quad y = m + \ell\beta.$$

If both are nonintegral, then $k, \ell > 0$, and

$$xy = nm + (n\ell + mk)\beta + k\ell\beta^2.$$

Membership in $\mathcal{S}$ would make this equal to $r + s\beta$, producing a nonzero quadratic polynomial over $\mathbb{Q}$ that vanishes at the transcendental number β . Equivalently, apply [theorem 5.1](#) to $(n + kT)(m + \ell T)$. Both are impossible. $\square$

Corollary 5.4 (Finite products). *[paper proof in this report] Let $x_1, \dots, x_r \in \mathcal{S}$ be positive. Then*

$$\prod_{i=1}^r x_i \in \mathcal{S} \iff \text{at most one } x_i \text{ is nonintegral.}$$

If zero factors are allowed, a zero product is of course a solution.

Proof. Under failure, after removing positive integral factors, the product is a polynomial in β . Its degree equals the number of noninteger factors and its leading coefficient is positive. A degree at least two is excluded by [theorem 5.1](#); degree zero or one is preserved by natural dilation. $\square$

Corollary 5.5 (Internal power test). *[paper proof in this report] For $x \in \mathcal{S}$ and every integer $r \geq 2$,*

$$x^r \in \mathcal{S} \iff x \in \mathbb{N}_0.$$

In particular, $x \in \mathbb{N}_0$ exactly when $x^2 \in \mathcal{S}$.

5.3 Exact quotient criterion

The same evaluation argument classifies division, which is more delicate because even the true solution set $\mathbb{N}_0$ is not closed under arbitrary quotients.

Theorem 5.6 (Quotients of solutions). *[paper proof in this report] Assume failure and write*

$$x = n + k\beta, \quad y = m + \ell\beta > 0.$$

Then $x/y \in \mathcal{S}$ if and only if exactly one of the following applicable conditions holds:

- (a) $\ell = 0$, so $y = m \in \mathbb{N}_{>0}$, and $m \mid n$ and $m \mid k$;
- (b) $\ell > 0$, and $x = qy$ for some $q \in \mathbb{N}_0$, equivalently $n = qm$ and $k = q\ell$.

In particular, if x, y are both noninteger, then

$$\frac{x}{y} \in \mathcal{S} \iff \frac{x}{y} \in \mathbb{N}_0.$$

Proof. Suppose $x/y = r + s\beta$ with $r, s \in \mathbb{N}_0$. Clearing the denominator gives

$$n + k\beta = mr + (ms + \ell r)\beta + \ell s\beta^2.$$

Transcendence forces $\ell s = 0$ and equality of the remaining coefficients. If $\ell = 0$, then $m > 0$ and the equations are $n = mr$, $k = ms$. If $\ell > 0$, then $s = 0$ and $n = mr$, $k = \ell r$, so $x = ry$. The converses follow by direct substitution. $\square$

A useful special case is

$$x > 0, \frac{1}{x} \in \mathcal{S} \iff x = 1.$$

Thus a hypothetical solution monoid is additively rich but almost maximally hostile to multiplication, powers, reciprocals, and quotients.

5.4 Closure formulations of the conjecture

Theorem 5.7 (Multiplicative closure equivalences). *[paper proof in this report] The following statements are equivalent.*

- (i) *The Alaoglu–Erdős conjecture is true.*
- (ii) *$\mathcal{S}$ is closed under multiplication.*
- (iii) *$\mathcal{S}$ is closed under squaring.*
- (iv) *For every real x ,*

$$2^x, 3^x \in \mathbb{Z} \implies 2^{x^2}, 3^{x^2} \in \mathbb{Z}.$$

Proof. If the conjecture holds, $\mathcal{S} = \mathbb{N}_0$, which has the stated closure properties. Multiplicative closure implies square closure. Conditions (iii) and (iv) are the same statement, and square closure plus [theorem 5.5](#) forces every solution to be integral. $\square$

For positive reals define the commutative operation

$$m \star n := 2^{(\log_2 m)(\log_2 n)} = m^{\log_2 n} = n^{\log_2 m}.$$

If $m = 2^x$ and $n = 2^y$, then $m \star n = 2^{xy}$.

Corollary 5.8 (Candidate star-product). [\[paper proof in this report\]](#) For $m, n \in \mathcal{C}$,

$$m \star n \in \mathcal{C} \iff m \text{ is a power of 2 or } n \text{ is a power of 2.}$$

The conjecture is therefore equivalent both to closure of $\mathcal{C}$ under $\star$ and to the apparently weaker diagonal condition

$$m \in \mathcal{C} \implies m \star m \in \mathcal{C}.$$

These are genuine equivalences, not a proof: there is presently no independent mechanism forcing the nonlinear closure. Their value is that they identify the missing property as a single crisp failure of semiring-like structure.

5.5 Lean formalization cost

The multiplication, quotient, and power criteria should be inexpensive to formalize from `PrimitiveGenerator.lean` and `Transcendence.lean`. A direct route avoids a general rational-function library:

1. split on `AlaogluErdosConjecture`;
2. in the failure branch obtain the canonical generator and unique coordinates;
3. expand the relevant equality as a polynomial of degree at most two in β ;
4. invoke transcendence to show all rational coefficients vanish;
5. discharge the remaining natural-number divisibility conditions with `omega`.

The full rational-function theorem needs evaluation injectivity over $\overline{\mathbb{Q}}$, but the core multiplication theorem only needs the already formalized transcendence over $\mathbb{Q}$.

6 Three-smooth outputs and transcendental product exponents

The multiplication theorem above uses the exact monoid and is therefore close to Lean. A second argument based on six exponentials gives more: when nonlinear closure fails, the relevant exponential values are usually not merely nonintegral but transcendental.

6.1 The two outputs cannot both be three-smooth

Theorem 6.1 (Three-smooth output test). [\[paper proof in this report\]](#) For every $x \in \mathcal{S}$, the following are equivalent:

- (i) $x \in \mathbb{N}_0$;
- (ii) both 2^x and 3^x are three-smooth;
- (iii) 6^x is three-smooth.

Proof. Condition (i) immediately implies (ii), and (ii) is equivalent to (iii) because $2^x, 3^x$ are positive integers and a product is three-smooth exactly when both factors are.

For (ii) $\Rightarrow$ (i), write

$$2^x = 2^a 3^b, \quad 3^x = 2^c 3^d$$

with $a, b, c, d \in \mathbb{N}_0$. Taking logarithms gives

$$x = a + b\theta, \quad x = d + \frac{c}{\theta}.$$

Therefore

$$b\theta^2 + (a - d)\theta - c = 0.$$

The repository proves θ transcendental, so every rational coefficient vanishes. Thus $b = c = 0$, $a = d$, and $x = a \in \mathbb{N}_0$. $\square$

This elementary-looking theorem uses the transcendence of θ , but not the exact conditional generator. It is unconditional and should be straightforward to add to the Lean development.

Corollary 6.2 (Primitive-core trichotomy). *[paper proof in this report] In the simultaneous least-output normalization*

$$M = 2^a w^d, \quad A = 3^b v^e,$$

at most one of $w = 3$ and $v = 2$ can hold. Equivalently, exactly one of the following three smoothness patterns occurs under failure:

M	A	primitive cores
three-smooth	not three-smooth	$w = 3, v \neq 2$
not three-smooth	three-smooth	$w \neq 3, v = 2$
not three-smooth	not three-smooth	$w \neq 3, v \neq 2$

The missing fourth pattern is impossible.

Indeed an odd, three-smooth, power-primitive integer greater than one is exactly 3, and a three-free, three-smooth, power-primitive integer greater than one is exactly 2.

6.2 A sharp third-base dichotomy from six exponentials

We use a standard form of the six-exponentials theorem [8, 11]: if u_1, u_2 are linearly independent over $\mathbb{Q}$ and v_1, v_2, v_3 are linearly independent over $\mathbb{Q}$, then at least one of the six numbers $e^{u_i v_j}$ is transcendental.

Lemma 6.3 (Third-base transcendence). *[classical theorem used] Let $y \in \mathcal{S} \setminus \mathbb{N}_0$, and let $B \in \mathbb{N}_{>0}$ be not three-smooth. Then B^y is transcendental.*

Proof. The three logarithms $\log 2, \log 3, \log B$ are linearly independent over $\mathbb{Q}$: a rational relation would give a multiplicative relation among $2, 3, B$, impossible because B has a prime factor outside $\{2, 3\}$. Also $1, y$ are linearly independent over $\mathbb{Q}$, since a rational solution is integral. Apply six exponentials. Five of the six values

$$2, 3, B, 2^y, 3^y, B^y$$

are algebraic—indeed, the first five are integers. Therefore B^y is transcendental. $\square$

Together with theorem 3.3, this gives an exact algebraicity trichotomy for integer bases:

$$B^y \in \overline{\mathbb{Q}} \iff B^y \in \mathbb{Z} \iff B \in \mathcal{H}_{2,3} \quad (y \in \mathcal{S} \setminus \mathbb{N}_0).$$

There is no algebraic-irrational middle case.

6.3 Products of noninteger solutions

Theorem 6.4 (Product-exponent dichotomy). *[classical theorem used] Let $x, y \in \mathcal{S}$.*

- (a) *If $x \in \mathbb{N}_0$ or $y \in \mathbb{N}_0$, then $2^{xy}, 3^{xy} \in \mathbb{N}$.*
- (b) *If both x and y are nonintegral, then each of 2^{xy} and 3^{xy} is either an integer or transcendental, at least one is transcendental, and*

$$\boxed{6^{xy} \text{ is transcendental.}}$$

Proof. Part (a) is natural-number dilation of a solution. For part (b), put $M_x = 2^x$ and $A_x = 3^x$. Then

$$2^{xy} = M_x^y, \quad 3^{xy} = A_x^y.$$

If M_x is three-smooth, [theorem 3.3](#) makes M_x^y an integer; if not, [theorem 6.3](#) makes it transcendental. The same holds for A_x^y . By [theorem 6.1](#), M_x, A_x cannot both be three-smooth, so at least one value is transcendental.

Finally $6^x = M_x A_x$ is an integer that is not three-smooth. Applying [theorem 6.3](#) with base 6^x and exponent y gives $(6^x)^y = 6^{xy}$ transcendental. $\square$

Corollary 6.5 (Diagonal transcendence). *[classical theorem used] If $x \in \mathcal{S} \setminus \mathbb{N}_0$, then 6^{x^2} is transcendental, and at least one of $2^{x^2}, 3^{x^2}$ is transcendental. More precisely,*

$$\begin{aligned} 2^{x^2} \in \mathbb{N} &\iff 2^x \in \mathcal{H}_{2,3}, \\ 3^{x^2} \in \mathbb{N} &\iff 3^x \in \mathcal{H}_{2,3}, \end{aligned}$$

and failure of either condition makes the corresponding value transcendental.

Corollary 6.6 (Mixed three-smooth bases). *[classical theorem used] Let $x, y \in \mathcal{S} \setminus \mathbb{N}_0$, and let $u, v \geq 1$. Then*

$$(2^u 3^v)^{xy}$$

is transcendental.

Proof. The integer

$$(2^u 3^v)^x = (2^x)^u (3^x)^v$$

has every prime occurring in either output. At least one output has a prime outside $\{2, 3\}$, so this base is not three-smooth. Apply [theorem 6.3](#) with exponent y . $\square$

6.4 A global second-level trichotomy

The prime-support theorem in [section 9](#) makes the preceding dichotomy uniform over all noninteger solutions, not dependent on a particular pair.

Theorem 6.7 (Uniform second-level pattern). *[classical theorem used] Assume failure and let $M = 2^\beta, A = 3^\beta$. For every $x, y \in \mathcal{S} \setminus \mathbb{N}_0$, exactly one of the following three global patterns holds:*

condition on (M, A)	2^{xy}	3^{xy}
$M \in \mathcal{H}_{2,3}, A \notin \mathcal{H}_{2,3}$	integer	transcendental
$M \notin \mathcal{H}_{2,3}, A \in \mathcal{H}_{2,3}$	transcendental	integer
$M \notin \mathcal{H}_{2,3}, A \notin \mathcal{H}_{2,3}$	transcendental	transcendental

The chosen pattern is independent of x and y .

Proof. Write $x = n + k\beta$ with $k > 0$. Then

$$2^x = 2^n M^k, \quad 3^x = 3^n A^k.$$

Multiplication by a power of the distinguished smooth prime cannot remove an outside prime, so 2^x is three-smooth exactly when M is; similarly for A . Now apply [theorem 6.4](#). The impossible fourth pattern is excluded by [theorem 6.1](#). $\square$

This theorem suggests a clean three-case research split. The one-smooth-output cases reduce to the exceptional primitive cores $w = 3$ or $v = 2$; the generic case has both second iterates transcendental.

6.5 Algebraicity formulations equivalent to the conjecture

Theorem 6.8 (Algebraicity tests). *[classical theorem used] Each of the following is equivalent to the Alaoglu–Erdős conjecture.*

- (i) *For every $x \in \mathcal{S}$, 6^{x^2} is algebraic.*
- (ii) *For every $x \in \mathcal{S}$, both 2^{x^2} and 3^{x^2} are algebraic.*
- (iii) *For every $x, y \in \mathcal{S}$, 6^{xy} is algebraic.*
- (iv) *For one fixed pair $u, v \geq 1$, every $x \in \mathcal{S}$ satisfies $(2^u 3^v)^{x^2} \in \overline{\mathbb{Q}}$.*
- (v) *For every $x \in \mathcal{S}$, the integer 6^x is three-smooth.*

Proof. If the conjecture is true, all relevant exponents are nonnegative integers, and all listed values are integers; condition (v) is also immediate. Conversely, a noninteger solution contradicts (i)–(iv) by [theorems 6.5](#) and [6.6](#), and contradicts (v) by [theorem 6.1](#). $\square$

These formulations replace integrality by the apparently weaker target of algebraicity. Six exponentials makes them equivalent. The exact missing step is now visible: one would need an independent reason that some nonlinear iterate such as 6^{x^2} is algebraic from the original two integer values.

6.6 Formalization boundary

The output-smooth theorem and the integer/noninteger parts of the product trichotomy use only the already formalized transcendence of θ , unique factorization, and [theorem 3.3](#). The transcendence upgrade requires a general six-exponentials theorem. The current repository determinant specializes to integer entries and proves the three-smooth classification. A promising formal route is to replace the determinant lower bound $|D| \geq 1$ by a number-field norm bound for a nonzero algebraic determinant. This would expose the same interpolation upper bound through an “algebraic entries of bounded height” interface and recover [theorem 6.3](#).

7 Common perfect-power content and exact affine descent

Assume failure throughout this section and fix the canonical generator β , with

$$M = 2^\beta, \quad A = 3^\beta.$$

For $x = n + k\beta$, exact coordinates give

$$2^x = 2^n M^k, \quad 3^x = 3^n A^k. \tag{4}$$

The coordinate gcd is an intrinsic arithmetic invariant of this output pair.

7.1 Simultaneous perfect powers

Theorem 7.1 (Exact common-power criterion). *[paper proof in this report] Let $x = n + k\beta \in \mathcal{S}$, and let $r \geq 1$. The following are equivalent:*

- (i) *both 2^x and 3^x are perfect r -th powers in $\mathbb{N}$;*
- (ii) *$x/r \in \mathcal{S}$;*
- (iii) *$r \mid n$ and $r \mid k$.*

Proof. If $2^x = U^r$ and $3^x = V^r$ with $U, V > 0$, uniqueness of positive real r -th roots gives

$$2^{x/r} = U, \quad 3^{x/r} = V,$$

so $x/r \in \mathcal{S}$. The converse follows by raising the two integer values at x/r to the r -th power.

Finally write $x/r = p + q\beta$ in unique coordinates. Then

$$n + k\beta = rp + rq\beta,$$

so $n = rp$, $k = rq$. This is exactly the pair of divisibility conditions, and the reverse construction is immediate. $\square$

For $x > 0$, define the *common output power degree*

$$g(x) := \max\{r \geq 1 : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\}.$$

Corollary 7.2 (Coordinate gcd). *[paper proof in this report] If $x = n + k\beta > 0$, then*

$$g(x) = \gcd(n, k).$$

The output pair is simultaneously power-primitive exactly when $\gcd(n, k) = 1$.

Thus every nonzero output pair has a unique decomposition

$$(2^x, 3^x) = (U^g, V^g), \quad g = \gcd(n, k),$$

where (U, V) is a simultaneously power-primitive output pair belonging to the solution x/g . The repository proves affine primitivity for the least pair; this corollary extends the intrinsic power degree to every point of the solution monoid.

7.2 Matched affine decompositions

A useful descent pattern strips the same distinguished-base depth and the same outer power from both outputs.

Theorem 7.3 (Matched affine decomposition criterion). *[paper proof in this report] Let $x = n + k\beta \in \mathcal{S}$, $d \geq 1$, and $r \in \mathbb{N}_0$. The following are equivalent:*

- (i) *there exist $U, V \in \mathbb{N}_{>0}$ such that*

$$2^x = 2^r U^d, \quad 3^x = 3^r V^d;$$

- (ii) *$(x - r)/d \in \mathcal{S}$;*
- (iii) *$0 \leq r \leq n$, $d \mid k$, and $d \mid (n - r)$.*

When these conditions hold,

$$\frac{x-r}{d} = \frac{n-r}{d} + \frac{k}{d}\beta, \quad U = 2^{(x-r)/d}, \quad V = 3^{(x-r)/d}.$$

Proof. Conditions (i) and (ii) are equivalent by positive-root uniqueness after removing the common factor 2^r or 3^r . Expanding (ii) in unique coordinates gives

$$n + k\beta = r + d(p + q\beta)$$

for some $p, q \in \mathbb{N}_0$. Thus $n = r + dp$ and $k = dq$, which is exactly (iii). The same equations give the formulas and the converse. $\square$

For a noninteger solution $x = n + k\beta$ with $k > 0$, Euclidean division $n = qk + r$, $0 \leq r < k$, gives a canonical reduced affine presentation

$$\boxed{x = r + k(q + \beta)}, \tag{5}$$

and hence

$$2^x = 2^r(2^q M)^k, \quad 3^x = 3^r(3^q A)^k.$$

The normalization $0 \leq r < k$ makes the presentation unique. It separates the outer affine degree k from the reduced common base depth r .

7.3 Why this does not yet descend below β

If n, k have a common divisor, [theorem 7.1](#) descends to $x/\gcd(n, k)$. If a matched decomposition exists, [theorem 7.3](#) descends to $(x-r)/d$. But exact coordinates ensure that the descended point remains $p + q\beta$ with $p, q \geq 0$; it is below β only when $q = 0$, in which case it is integral. The least noninteger point β is already primitive with respect to every such descent. A successful descent argument therefore needs a transformation not captured by common powers and common distinguished-base depths.

8 Primitive-output statistics

The common-power criterion translates ordinary coprimality statistics of lattice points into exact statistics of hypothetical integer-output pairs.

8.1 Total and primitive counts

For $T > 0$, let

$$N_\beta(T) = \#\{(n, k) \in \mathbb{N}_0^2 \setminus \{(0, 0)\} : n + k\beta < T\}$$

and

$$P_\beta(T) = \#\{(n, k) \in \mathbb{N}_0^2 : n + k\beta < T, \gcd(n, k) = 1\}.$$

The first counts nonzero solutions below T ; the second counts output pairs with no nontrivial common perfect-power degree. Recall

$$N_\beta(T) = \frac{T^2}{2\beta} + O_\beta(T).$$

Theorem 8.1 (Primitive output pairs). *[paper proof in this report]* As $T \rightarrow \infty$,

$$P_\beta(T) = \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T) = \frac{3T^2}{\pi^2\beta} + O_\beta(T \log T).$$

Consequently,

$$\frac{P_\beta(T)}{N_\beta(T)} \longrightarrow \frac{1}{\zeta(2)} = \frac{6}{\pi^2}.$$

Proof. Every nonzero coordinate pair has the unique factorization

$$(n, k) = d(n_0, k_0), \quad d = \gcd(n, k), \quad \gcd(n_0, k_0) = 1.$$

Hence

$$N_\beta(T) = \sum_{d \geq 1} P_\beta(T/d),$$

and Möbius inversion gives

$$P_\beta(T) = \sum_{d \leq T} \mu(d) N_\beta(T/d).$$

The lattice estimate is uniform in the form $N_\beta(U) = U^2/(2\beta) + O_\beta(U + 1)$. Therefore

$$\begin{aligned} P_\beta(T) &= \frac{T^2}{2\beta} \sum_{d \leq T} \frac{\mu(d)}{d^2} + O_\beta\left(\sum_{d \leq T} \left(\frac{T}{d} + 1\right)\right) \\ &= \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T). \end{aligned}$$

Divide by the total asymptotic to obtain the limit. □

The universal constant $6/\pi^2$ is independent of the hypothetical generator. Geometry contributes $1/(2\beta)$; primitive lattice points contribute $1/\zeta(2)$.

8.2 Distribution of maximal common power degree

For fixed $g \geq 1$, let

$$P_{\beta,g}(T) := \#\{x = n + k\beta : 0 < x < T, \gcd(n, k) = g\}.$$

Corollary 8.2 (Exact-degree distribution). *[paper proof in this report]* For fixed $g \geq 1$,

$$P_{\beta,g}(T) = P_\beta(T/g) = \frac{3T^2}{\pi^2\beta g^2} + O_\beta\left(\frac{T}{g} \log \frac{2T}{g}\right).$$

The limiting probability that an output pair has maximal common power degree exactly g is

$$\frac{6}{\pi^2 g^2}.$$

The probabilities sum to one. Likewise, the two outputs are both perfect r -th powers exactly when $r \mid n, k$, so

$$\#\{0 < x < T : 2^x, 3^x \text{ both perfect } r\text{-th powers}\} = N_\beta(T/r), \tag{6}$$

and the limiting proportion is $1/r^2$.

8.3 Exact generating series

The coordinate monoid has the Laplace series

$$\begin{aligned}
 Z_\beta(s) &:= \sum_{x \in \mathcal{S} \setminus \{0\}} e^{-sx} \\
 &= \sum_{n,k \geq 0} e^{-s(n+k\beta)} - 1 \\
 &= \boxed{\frac{1}{(1-e^{-s})(1-e^{-\beta s})} - 1}, \quad s > 0.
 \end{aligned} \tag{7}$$

The primitive-output series is

$$Z_\beta^{\text{prim}}(s) = \sum_{d \geq 1} \mu(d) Z_\beta(ds).$$

At the candidate level one similarly has the Dirichlet series

$$\sum_{m \in \mathcal{C}} m^{-s} = \frac{1}{(1-2^{-s})(1-M^{-s})}, \quad \Re s > 0.$$

These identities make the two free generators visible and give clean starting points for weighted Tauberian refinements.

8.4 Interpretation

Roughly 60.79% of hypothetical output pairs would be simultaneously power-primitive. This is not pressure toward a contradiction: it is exactly the expected primitive-lattice proportion for a free rank-two monoid. Its value is diagnostic. Any descent argument must act on the primitive majority, not merely on the exceptional pairs with a common outer power.

9 Candidate divisibility and ambient root saturation

The candidate monoid also has a simple valuation geometry. It yields closure properties not visible from the defining condition $m^\theta \in \mathbb{N}$.

9.1 A two-dimensional valuation cone

Under failure write

$$M = 2^a U, \quad U > 1 \text{ odd.}$$

A candidate with coordinates (n, k) is

$$m(n, k) = 2^n M^k = 2^{n+ak} U^k.$$

Set $s = v_2(m) = n + ak$. The map

$$m(n, k) \mapsto (s, k)$$

is a bijection from $\mathcal{C}$ to

$$\Lambda_a := \{(s, k) \in \mathbb{N}_0^2 : s \geq ak\}.$$

Because U is odd and greater than one,

$$m(n, k) \mid m(n', k') \iff s \leq s' \text{ and } k \leq k'. \tag{8}$$

Thus candidate divisibility is coordinatewise order on a lattice cone.

Theorem 9.1 (Candidate gcd/lcm lattice). *[paper proof in this report] Unconditionally, $\mathcal{C}$ is closed under gcd and lcm. Under failure, if $m_i = m(n_i, k_i)$ and $s_i = n_i + ak_i$, then*

$$\begin{aligned}\gcd(m_1, m_2) &= m(\min(s_1, s_2) - a \min(k_1, k_2), \min(k_1, k_2)), \\ \text{lcm}(m_1, m_2) &= m(\max(s_1, s_2) - a \max(k_1, k_2), \max(k_1, k_2)).\end{aligned}$$

Candidate divisibility is therefore a distributive lattice.

Proof. If the conjecture is true, candidates are powers of two. Under failure, gcd and lcm take coordinatewise minima and maxima of the prime-valuation vectors of $2^s U^k$. The resulting points remain in Λ_a ; coordinatewise min and max form a distributive lattice. $\square$

Corollary 9.2 (Candidate divisors). *[paper proof in this report] The number of candidate divisors of $m(n, k)$, including 1 and the number itself, is*

$$(k+1)(n+1) + \frac{ak(k+1)}{2}.$$

Proof. A candidate divisor corresponds to $(s', j) \in \Lambda_a$ with $0 \leq j \leq k$ and $aj \leq s' \leq n + ak$. There are $n + ak - aj + 1$ choices for s' . Sum this arithmetic progression over $j = 0, \dots, k$. $\square$

The gcd/lcm closure is compatible with failure, unlike the nonlinear star-product closure. It is therefore structural information rather than a direct proof route.

9.2 Fixed prime support

Use the simultaneous normalization

$$M = 2^a w^d, \quad A = 3^b v^e.$$

For $x = n + k\beta$,

$$2^x = 2^{n+ak} w^{dk}, \quad 3^x = 3^{n+bk} v^{ek}.$$

Corollary 9.3 (Prime-support rigidity). *[paper proof in this report] Every noninteger solution ($k > 0$) has exactly the same odd prime support in its base-2 output, namely $\text{supp}(w)$, and exactly the same prime support away from 3 in its base-3 output, namely $\text{supp}(v)$. The only support variation is whether the distinguished primes 2 and 3 occur.*

A counterexample would therefore lock the entire noninteger branch onto one fixed pair of finite prime sets. Computational searches should exploit this by enumerating primitive supports or cores, not raw integers independently.

9.3 Ambient root saturation and the index d

Use the canonical form $M = 2^a w^d$, where $w > 1$ is odd and the gcd of its positive prime valuations is one. Reparameterizing by the exponent of w gives

$$\mathcal{C} = \{2^s w^t : s, t \in \mathbb{N}_0, d \mid t, ds \geq at\}. \quad (9)$$

For a multiplicative monoid $H \subseteq \mathbb{N}_{>0}$, define its *ambient root saturation*

$$\text{Sat}(H) := \{u \in \mathbb{N}_{>0} : u^r \in H \text{ for some } r \geq 1\}.$$

Theorem 9.4 (Exact ambient root saturation). *[paper proof in this report] Under failure,*

$$\text{Sat}(\mathcal{C}) = \{2^s w^t : s, t \in \mathbb{N}_0, ds \geq at\}.$$

For $u = 2^s w^t$ in the saturation, the least positive r for which $u^r \in \mathcal{C}$ is

$$r_{\min}(u) = \frac{d}{\gcd(d, t)}.$$

Proof. Suppose first that $u = 2^s w^t$ and $ds \geq at$. Put $r = d / \gcd(d, t)$. Then $d \mid rt$, and scaling the cone inequality gives $d(rs) \geq a(rt)$. Hence $u^r \in \mathcal{C}$. The divisibility condition $d \mid rt$ proves minimality.

Conversely suppose $u^r \in \mathcal{C}$, so

$$u^r = 2^S w^{dk}$$

with $dS \geq adk$. No prime outside $\{2\} \cup \text{supp}(w)$ divides u . Write $c_p = v_p(w)$. For every $p \mid w$,

$$rv_p(u) = dk c_p.$$

Since $\gcd\{c_p : p \mid w\} = 1$, Bézout gives $dk/r \in \mathbb{Z}$; call it t . The odd part of u is then w^t , while $S = rs$ with $s = v_2(u)$. Substitution in the cone inequality yields $ds \geq at$. $\square$

Corollary 9.5 (Ambient saturation index). *[paper proof in this report] The candidate monoid equals its ambient root saturation exactly when $d = 1$. More generally,*

$$d = \min\{r \geq 1 : \exists s \in \mathbb{N}_0, (2^s w)^r \in \mathcal{C}\}.$$

A terminology caution is important. In standard affine-semigroup theory, normality is measured relative to the group generated by the semigroup. Here that group already has $t \equiv 0 \pmod{d}$, and $\mathcal{C}$ is normal in that intrinsic lattice for every d . The theorem instead measures saturation in the larger ambient lattice containing every $2^s w^t$. Thus d is a finite-index *over-lattice saturation defect*, not a failure of intrinsic semigroup normality. This is precisely parallel to its field-theoretic meaning as the least rational-power index and exact radical degree.

10 Optimizing the higher-difference obstruction

The arbitrary-order theorem in `HigherDifference.lean` permits the order to grow with the initial candidate size. Optimizing that order strengthens every fixed-order corollary, but also quantifies why the method does not presently contradict the conditional monoid.

10.1 The exact derivative coefficient

For $r \geq 2$ and $1 < \theta < 2$, set

$$c_r = |\theta(\theta - 1) \cdots (\theta - r + 1)|.$$

Because the factors from $2 - \theta$ onward are positive after taking absolute values,

$$c_r = \frac{\theta(\theta - 1)}{\Gamma(2 - \theta)} \Gamma(r - \theta). \quad (10)$$

Stirling’s formula gives

$$c_r^{1/r} = \frac{r}{e} \left(1 + O\left(\frac{\log r}{r}\right) \right). \quad (11)$$

The formalized progression theorem applies whenever

$$h < c_r^{-1/r} n^{1-\theta/r}.$$

Using (11), the right side is

$$\frac{en}{r} \exp\left(-\frac{\theta \log n}{r}\right) \left(1 + O\left(\frac{\log r}{r}\right)\right). \quad (12)$$

Ignoring the lower-order factor, the logarithmic derivative with respect to r is

$$-\frac{1}{r} + \frac{\theta \log n}{r^2},$$

so the maximum occurs at $r \sim \theta \log n$. At that value the explicit factor e and the exponential e^{-1} cancel.

Theorem 10.1 (Optimized variable-order obstruction). *[paper proof in this report] Let $0 < \varepsilon < 1$. For all sufficiently large n , choose an integer*

$$r_n = \theta \log n + O(1), \quad r_n \geq 2.$$

Then candidates cannot occupy all terms of

$$n, n+h, \dots, n+r_n h$$

whenever

$$1 \leq h \leq (1-\varepsilon) \frac{n}{\theta \log n}.$$

Moreover, the largest leading-order step scale obtainable by optimizing the existing curvature condition over r is

$$h_{\max}(n) \sim \frac{n}{\theta \log n}.$$

Proof. Choose r_n to be either nearest integer to $\theta \log n$. Then $r_n \rightarrow \infty$, so (12) applies and its quotient by $n/(\theta \log n)$ tends to one. The fixed $(1-\varepsilon)$ margin absorbs the Stirling error and the $O(1)$ integer rounding. The formalized condition (3) then excludes the progression.

For the optimality assertion, the leading approximation in (12) is a unimodal function of positive real r , with its unique maximum at $\theta \log n$. Regimes where r remains bounded or grows outside a constant multiple of $\log n$ give a strictly smaller order or leading constant. $\square$

10.2 Comparison with conditional candidate spacing

A dyadic block $[X, 2X)$ is an interval of length one in exponent space. Exact coordinates give

$$\#(\mathcal{C} \cap [X, 2X)) = \frac{\log X}{\beta \log 2} + O_\beta(1). \quad (13)$$

The average arithmetic spacing in the block is therefore

$$\sim \beta \log 2 \frac{X}{\log X}.$$

At $n \asymp X$, the optimized curvature threshold is

$$\sim \frac{1}{\theta} \frac{X}{\log X}.$$

The ratio is

$$\frac{\beta \log 2}{1/\theta} = \boxed{\beta \log 3}. \quad (14)$$

The kernel-verified bound $\beta \geq 12$ already makes this ratio greater than 13.18; the software-level $\beta > 26$ bound makes it greater than 28.56.

The same constant appears in a cardinality comparison. The forbidden progression would require

$$r_n + 1 \sim \theta \log X$$

candidates in one structured configuration, while the entire dyadic block contains only

$$\sim \frac{\log X}{\beta \log 2}$$

candidates. Their ratio is again $\beta \log 3$.

Remark 10.2 (Quantified barrier). The variable-order optimization materially strengthens the local theorem, but total candidate density is too small by a fixed factor of at least $\beta \log 3$. No argument using only the number of candidates in a dyadic block can force the progression required by the current finite-difference functional. A successful continuation must exploit additional arithmetic correlation—for example, valuation congruences that force a special progression—or find a different integer-valued functional with a better curvature-to-spacing constant.

This is a limitation of the method, not evidence against the conjecture. Its value is to replace the vague statement “density seems too weak” by a sharp asymptotic constant and a concrete target for any improved functional.

11 A directed-rounding exclusion through 2^{26}

This section records a new finite computation and its exact trust boundary. It is not a substitute for Lean: its purpose is to strengthen the explicit zero-free region, test the conditional models, and provide data for future certificate generation.

11.1 What must be certified for each integer

For a non-power of two m , let

$$y = m^\theta, \quad \theta = \frac{\log 3}{\log 2}.$$

To prove that $y \notin \mathbb{N}$, it suffices to find $n \in \mathbb{N}$ and certify the strict pair

$$n < m^\theta < n + 1.$$

Avoiding direct exponentiation, these are equivalent to

$$\log m \log 3 - \log n \log 2 > 0, \quad (15)$$

$$\log(n + 1) \log 2 - \log m \log 3 > 0. \quad (16)$$

All logarithms and products are positive in the scanned range, so interval directions are simple and monotone.

The verifier uses ordinary `long double` only to propose $n \approx \lfloor m^\theta \rfloor$. It tests the offsets

$$0, -1, 1, -2, 2, -3, 3.$$

A proposal is accepted only when both (15) and (16) have a strictly positive *directed lower bound* computed by MPFR. Failure to locate an interval is reported as a failure, never silently classified as nonintegral. Thus locator error can create a false alarm but cannot create a false proof.

11.2 Directed enclosures

At precision $p = 192$ bits, the program constructs

$$\begin{aligned} L_m^- &\leq \log m \leq L_m^+, & L_2^- &\leq \log 2 \leq L_2^+, & L_3^- &\leq \log 3 \leq L_3^+, \\ L_n^+ &\geq \log n, & L_{n+1}^- &\leq \log(n+1). \end{aligned}$$

It then rounds downward in

$$\delta_- = L_m^- L_3^- - L_n^+ L_2^+, \quad \delta_+ = L_{n+1}^- L_2^- - L_m^+ L_3^+.$$

Thus $\delta_- > 0$ proves (15), and $\delta_+ > 0$ proves (16). The input integers are below 2^{42} on the n -side and are exactly representable at 192-bit precision.

Powers of two are skipped, not tested: if $m = 2^q$, then $m^\theta = 3^q$ is the known integral candidate. Every other positive integer is tested. An exact integral value at a non-power of two could satisfy neither strict interval for any n , so it would necessarily be reported as a failure.

11.3 Execution environment and source identity

The recorded run used:

- MPFR 4.2.2, linked as `libmpfr.so.6`;
- GCC 14.2.0 with `-O3 -fopenmp`;
- GNU OpenMP with five worker threads;
- Linux x86-64;
- 192-bit MPFR precision and directed modes `MPFR_RNDD/MPFR_RNDU`.

The exact executed C source is reproduced in [section B](#). Its SHA-256 digest is

f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d

The execution image contained the MPFR shared library but not the development header, so the source has a guarded minimal declaration of the MPFR 4.x ABI. On an ordinary system with `mpfr.h`, the standard header branch is used. This ABI detail is part of the software trust boundary and is disclosed rather than hidden.

11.4 Result

There are $2^{26} - 1 = 67,108,863$ positive integers below 2^{26} . Exactly 26 are powers of two, from 2^0 through 2^{25} . Hence the verifier tests

$$67,108,863 - 26 = \boxed{67,108,837}$$

non-powers. The range was divided at multiples of 2^{22} into one initial chunk $[1, 2^{22})$ and fifteen chunks of width 2^{22} ; all chunks returned zero failures. The complete chunk audit appears in [section C](#).

Theorem 11.1 (Software-level finite exclusion). *[directed-rounding computation] For every integer m with $1 \leq m < 2^{26}$,*

$$m^\theta \in \mathbb{N} \implies m \text{ is a power of } 2.$$

Consequently, at the stated software trust level,

$$x \in \mathcal{S} \setminus \mathbb{N}_0 \implies \boxed{x > 26}.$$

Explanation of the consequence. The verifier establishes the first implication for every non-power input; powers of two are the known candidates. If x were nonintegral, then $m = 2^x$ would be an integer candidate but not a power of two. The finite exclusion gives $m \geq 2^{26}$. Equality is itself a power of two and would force $x = 26$, so in fact $m > 2^{26}$, hence $x > 26$. $\square$

11.5 Closest approaches

The smallest certified lower logarithmic margin over the full scan is

$$3.6447144482202515 \times 10^{-20}$$

at

$$(m, n) = (58,570,438, 2,048,703,434,093).$$

A separate 100-decimal evaluation gives

$$m^\theta - n = 1.07725158750919017548 \dots \times 10^{-7}.$$

The smallest certified upper logarithmic margin is

$$5.4132120119800561 \times 10^{-21}$$

at

$$(m, n) = (29,411,704, 687,581,893,443),$$

where

$$n + 1 - m^\theta = 5.36974926710988874953 \dots \times 10^{-9}.$$

The same logarithmic upper margin recurs at the scaled point $(58,823,408, 2,062,745,680,331)$, as expected from multiplication of m by two and the identity $(2m)^\theta = 3m^\theta$.

Both extremal inequalities were independently reevaluated at 100 decimal digits. This secondary check is not an independent proof system, but it is a useful guard against transcription and aggregation errors. The certified margins are many orders of magnitude larger than the nominal 192-bit rounding scale of the logarithmic operations.

11.6 Trust boundary and route to a kernel theorem

The computation trusts the C source, GCC code generation, OpenMP scheduling, the MPFR and GMP shared libraries, the x86-64 arithmetic environment, and the operating system. MPFR’s correct-rounding contract makes the numerical reasoning crisp, but it does not remove those software assumptions.

The natural next step is certificate extraction rather than a still larger blind scan. For each block, one can record rational lower and upper approximants to θ and integer inequalities sufficient to replay every exclusion. Compression should exploit monotonicity and convexity so that one certificate covers a run of consecutive m , rather than storing one record per input. The existing 4096 Lean proof provides the checker pattern. A practical intermediate target is 2^{16} , followed by a generated hierarchical certificate through 2^{26} .

12 A consolidated catalogue of equivalent formulations

The repository already contains many equivalences: candidate powers of two, odd localized radicals, primitive radicals, least-generator nonexistence, and subquadratic counting. The new results add nonlinear closure and algebraicity tests. The following theorem collects the most useful forms in one place.

Theorem 12.1 (Equivalent targets). *The following assertions are equivalent.*

- (1) [\[kernel-verified Lean\]](#) The Alaoglu–Erdős conjecture: $\mathcal{S} = \mathbb{N}_0$.
- (2) [\[kernel-verified Lean\]](#) Every candidate is a power of two.
- (3) [\[kernel-verified Lean\]](#) There is no odd $u > 1$ with $u^\theta \in \mathbb{Z}[1/3]$; equivalently, none with u power-primitive.
- (4) [\[paper proof in this report\]](#) $\mathcal{S}$ is closed under multiplication.
- (5) [\[paper proof in this report\]](#) $\mathcal{S}$ is closed under squaring.
- (6) [\[paper proof in this report\]](#) $\mathcal{C}$ is closed under the operation $m \star n = 2^{\log_2 m \log_2 n}$.
- (7) [\[paper proof in this report\]](#) Every $m \in \mathcal{C}$ satisfies $m \star m \in \mathcal{C}$.
- (8) [\[paper proof in this report\]](#) Every $x \in \mathcal{S}$ has both outputs $2^x, 3^x$ three-smooth.
- (9) [\[paper proof in this report\]](#) Every $x \in \mathcal{S}$ has 6^x three-smooth.
- (10) [\[classical theorem used\]](#) Every $x \in \mathcal{S}$ has $6^{x^2} \in \overline{\mathbb{Q}}$.
- (11) [\[classical theorem used\]](#) Every $x \in \mathcal{S}$ has both $2^{x^2}, 3^{x^2} \in \overline{\mathbb{Q}}$.
- (12) [\[classical theorem used\]](#) Every $x, y \in \mathcal{S}$ has $6^{xy} \in \overline{\mathbb{Q}}$.
- (13) [\[kernel-verified Lean\]](#) The candidate count below 2^T is $o(T^2)$.

Proof. The repository proves the equivalence of (1)–(3) and (13). The equivalence of (1) with (4)–(7) is [theorems 5.7 and 5.8](#). The equivalence with (8)–(9) is [theorem 6.1](#). The equivalence with (10)–(12) is [theorem 6.8](#). $\square$

The formulations are not equally promising. Counting is very far from the final integrality statement; multiplication closure is exact but has no known independent source; the radical formulation is closest to established transcendence theory; and the three-smooth/algebraicity formulations offer a possible route through a third-base theorem.

12.1 A third-base manufacturing principle

[Theorem 3.3](#) suggests a particularly concrete target. Given a hypothetical noninteger β , it would suffice to construct *one* positive integer $B \notin \mathcal{H}_{2,3}$ such that $B^\beta \in \mathbb{Z}$. This would immediately contradict the formalized three-smooth-base theorem.

The obvious choices are $B = M = 2^\beta$, $B = A = 3^\beta$, or $B = MA = 6^\beta$. Their β -th powers are

$$2^{\beta^2}, \quad 3^{\beta^2}, \quad 6^{\beta^2}.$$

Thus “manufacture a third base” is exactly the square-closure problem in a different language. The six-exponentials result makes the stakes sharper: whenever the chosen base is not three-smooth, its β -th power is already known to be transcendental. Any proof of algebraicity would therefore finish the conjecture at once.

This framing is useful because it isolates the missing bridge:

derive one nonlinear algebraic value from the two linear integer values.

Neither unique factorization, ordinary exponent laws, nor linear forms in logarithms currently supply that bridge.

13 Prioritized future research program

The following agenda is ordered by a combination of mathematical leverage, falsifiability, and formalization value. It is intentionally technical rather than promotional.

13.1 Priority 1: formalize the new elementary rigidity theorems

[research direction] Add Lean modules for:

- three-smooth output equivalence, [theorem 6.1](#);
- exact multiplication and power criteria, [theorems 5.3](#) and [5.5](#);
- quotient classification, [theorem 5.6](#);
- common perfect-power and affine-decomposition criteria, [theorems 7.1](#) and [7.3](#);
- candidate gcd/lcm closure and divisor enumeration, [theorems 9.1](#) and [9.2](#);
- ambient root saturation, with the intrinsic-lattice terminology caveat;
- primitive lattice asymptotics, initially as exact finite Möbius identities and later through Mathlib asymptotics.

These proofs are short relative to the existing determinant and Gelfond–Schneider machinery. Formalizing them would give the project a larger library of reusable “counterexample-world eliminators” and guard against subtle sign, divisibility, or normalization errors.

A suggested source split is:

module	contents
<code>RationalOperationRigidity.lean</code>	products, powers, quotients, specialized polynomial evaluation
<code>OutputSmoothness.lean</code>	three-smooth output test and primitive-core trichotomy
<code>CommonPowerContent.lean</code>	exact common power degree and affine descent
<code>CandidateDivisibility.lean</code>	gcd/lcm lattice, candidate divisors, support rigidity
<code>AmbientRootSaturation.lean</code>	exact over-lattice saturation and least root exponent
<code>PrimitiveOutputCounting.lean</code>	finite Möbius identities and asymptotic wrappers

13.2 Priority 2: turn the 2^{26} scan into replayable certificates

[research direction] The next computational objective should not simply be 2^{27} . It should be a compact certificate format accepted by a small Lean checker. Useful compression mechanisms include:

1. rational bounds $p/q < \theta < r/s$ proved once;
2. monotonicity blocks in which the same floor candidate is controlled by endpoint inequalities;
3. convexity bounds for m^θ over integer intervals;
4. fixed-point integer arithmetic after scaling logarithmic intervals;
5. a hierarchy of block certificates, allowing local rechecking and parallel generation.

A checker should verify the mathematical implication “certificate $\Rightarrow$ no integer value” with no floating-point operations. External MPFR code may generate the certificate but is then removed from the theorem’s trust base. The existing 4096 modules are the prototype; the open engineering problem is certificate compression.

13.3 Priority 3: exploit the uniform smoothness trichotomy

[research direction] Split every hypothetical counterexample into the three cases of [theorem 6.7](#).

Case I: the base-2 output is three-smooth. Then the primitive odd core is $w = 3$, so

$$\beta = a + d\theta, \quad M = 2^a 3^d,$$

and

$$A = 3^\beta = 3^a (3^\theta)^d \in \mathbb{N}.$$

Thus 3^θ is a positive radical whose d -th power lies in $\mathbb{Z}[1/3]$. This is the sharp exceptional target

$$3^{\log_2 3}.$$

The output normalization, exact radical degree, and gcd condition should be specialized to this one-core case; they may impose stronger congruence or ramification restrictions than are visible in the generic notation.

Case II: the base-3 output is three-smooth. Symmetrically the three-free primitive core is $v = 2$, and

$$\beta = b + \frac{e}{\theta}, \quad A = 3^b 2^e,$$

so $2^{1/\theta} = 2^{\log_3 2}$ is a positive radical of a rational with denominator supported at 2. This case deserves a formally symmetric normalization theorem; currently most of the repository is presented from the base-2 candidate side.

Case III: neither output is three-smooth. Then for every pair of noninteger solutions, both 2^{xy} and 3^{xy} are transcendental. This is the generic case. Its advantage is two independent outside-prime supports; a mixed Archimedean/non-Archimedean determinant may exploit both simultaneously.

The three cases have genuinely different arithmetic. Treating them separately is more promising than carrying a generic primitive core through every argument.

13.4 Priority 4: combine the Archimedean determinant with prime valuations

[research direction] The current determinant lower bound uses only that a nonzero integer has absolute value at least one. Candidate outputs contain much more information: their valuations at primes in $\text{supp}(w)$ and $\text{supp}(v)$ are exact linear functions of the coordinate k . A future determinant should be designed so that column or row differences force a large predictable divisor while interpolation still makes its Archimedean absolute value small.

A concrete experimental template is:

1. choose entries from $2^{n+k\beta} = 2^n M^k$ and $3^{n+k\beta} = 3^n A^k$, or from monomials in both;
2. arrange finite differences in k so that powers of M , A , or factors $M^h - 1$, $A^h - 1$ divide minors;
3. use lifting-the-exponent lemmas at primes from fixed supports to quantify the non-Archimedean gain;
4. normalize away trivial row factors before estimating the Archimedean determinant;
5. seek a product-formula contradiction rather than the elementary $|D| \geq 1$.

The danger is that the same large factors also enlarge the Archimedean determinant and cancel the gain. Small symbolic and numerical models should be used to calculate the exact balance before investing in formalization. The target is a genuine improvement over the three-by-two dimensional count, not a renamed form of the existing determinant.

13.5 Priority 5: local Kummer analysis of the exact radical

[research direction] The canonical radical

$$E = w^\theta, \quad E^d = A/3^a, \quad [\mathbb{Q}(E) : \mathbb{Q}] = d$$

produces a pure extension with an exact rational-power index. Apply local Kummer and binomial irreducibility theory prime by prime:

- classify possible ramification at primes dividing d , A , w , and 3;
- combine the power-primitivity of w with Capelli-type conditions on $X^d - A/3^a$;
- compare the splitting fields of E and the multiplicative relation $E = 3^{\log_2 w}$;
- search for a local obstruction in the exceptional cases $w = 3$ or $v = 2$;
- formalize exact denominator and valuation consequences even when the global transcendence step remains open.

Pure-extension theory alone is unlikely to prove the conjecture, because arbitrary positive rational radicals certainly exist. The extra datum is the logarithmic identity (1). The research question is whether local restrictions on the radical can be made incompatible with that specific Archimedean logarithmic origin.

13.6 Priority 6: generalize the determinant from integer to algebraic entries

[research direction] Formalize a six-exponentials theorem with algebraic, rather than merely integral, matrix entries. If the entries lie in a number field K , a nonzero determinant $D \in K$ can be bounded below through

$$1 \leq |N_{K/\mathbb{Q}}(\mathfrak{d}D)|$$

after clearing a controlled denominator $\mathfrak{d}$. The conjugates and heights then enter the upper bound. This is technically substantial, but it would:

- formalize [theorem 6.3](#) and all algebraicity equivalences;
- produce a reusable Mathlib-level six-exponentials component;
- clarify exactly which height losses prevent shrinking from three logarithms to two;
- create an interface for mixed algebraic auxiliary values, not only integers.

Even if this does not settle Alaoglu–Erdős, it would turn the current informal transcendence upgrade into a kernel-checked theorem and expose the quantitative boundary in a reusable form.

13.7 Priority 7: support-first and normalization-first computation

[research direction] Raw scanning treats each m independently. Under failure, every noninteger candidate has the form

$$m = 2^{n+ak} w^{dk}$$

for one fixed primitive support. A structured search should enumerate:

1. a primitive odd core w (valuation gcd one);

2. the least rational-power index d ;
3. the reduced depth a ;
4. the nearest integer A to $3^a w^{d\theta}$;
5. the simultaneous base-3 primitive normalization and gcd obstruction.

Cheap filters include parity, perfect-power primitivity, denominator support, the exact common-coordinate gcd condition, and the three-smooth-output trichotomy. For a bounded range this is equivalent to the raw search, but it groups inputs by the hypothetical global structure and may reveal forbidden support patterns. It also produces data aligned with the Lean theorems rather than a flat list of near misses.

13.8 Priority 8: seek arithmetic forcing for finite differences

[research direction] The optimized higher-difference theorem is already near its analytic limit. Progress now requires a mechanism that *forces* many candidates into a short additive pattern. Potential sources are:

- congruences among candidates with the same k -coordinate;
- gcd/lcm lattice rectangles whose ordinary integer combinations have controlled size;
- valuation pigeonhole principles inside a dyadic block;
- polynomial rather than arithmetic progressions, paired with a matching integer-valued difference operator;
- mixed differences in (n, k) -coordinates whose image under $m = 2^n M^k$ has stronger curvature constants.

Every proposal should be tested against the exact conditional monoid. If the desired pattern is not forced in a generic free monoid $\{2^n M^k\}$, then density alone will not supply it.

13.9 Priority 9: isolate a nonlinear algebraicity bridge

[research direction] By [theorem 6.8](#), any proof that 6^{β^2} is algebraic from the original hypotheses completes the conjecture. Equivalent targets are algebraicity of both $2^{\beta^2}, 3^{\beta^2}$, or integrality of one non-three-smooth base raised to β . Possible sources of such a bridge include:

- an additional functional equation extracted from leastness or primitive normalization;
- an algebraic relation among M, A forced by matching prime valuations;
- a determinant whose vanishing yields algebraicity of a second iterate;
- a rigidity theorem for algebraic values of $z \mapsto z^\beta$ on a finitely generated multiplicative group stronger than six exponentials.

This is the highest-risk direction but the one most directly aligned with the exact missing step. The closure and transcendence theorems in this report provide clean tests: any claimed bridge must contradict the conditional monoid rather than merely restate it.

14 Conclusions

The Alaoglu–Erdős conjecture survives this investigation. The existing Lean formalization already places a hypothetical counterexample in an extraordinarily narrow shape: a transcendental least generator, a unique primitive odd core, an exact radical degree, a free rank-two solution monoid,

fixed output supports, and strong local progression exclusions. None of these constraints alone is inconsistent.

The new results sharpen that picture in directions not present in the starting report. A counterexample world would have exact rational-function rigidity: the only rational expressions in the generator that return to the solution set are its natural affine combinations. Products of two noninteger solutions never return; quotients return only as natural multiples; common output powers are measured exactly by coordinate gcd; and candidate divisibility forms a distributive lattice with a computable ambient saturation. Six exponentials upgrades the nonlinear failure from nonintegrality to transcendence: 6^{xy} is transcendental for every pair of noninteger solutions.

These are meaningful advances, but they also make the final boundary cleaner. To prove the conjecture one must either:

1. derive one nonlinear algebraic value from $2^x, 3^x \in \mathbb{N}$, thereby contradicting the new transcendence dichotomy;
2. manufacture a non-three-smooth integer base with integral x -th power;
3. find a local valuation obstruction to the exact canonical radical; or
4. build a determinant or product-formula argument that uses integrality beyond the generic four-exponentials setup.

The finite computation moves the explicit software-level frontier from $x > 20$ to $x > 26$, while the variable-order analysis explains why finite differences need new arithmetic forcing. The most immediate durable gains are to formalize the elementary new theorems and convert the MPFR scan into replayable certificates. The most promising mathematical split is the uniform three-case classification according to which least outputs are three-smooth, with separate local and transcendence attacks in each case.

No hidden assumption in this report turns these partial results into a proof. What has improved is the map of the remaining territory: the conjecture is now surrounded by a dense set of exact, checkable equivalences and obstructions, and the missing ingredient can be stated with considerably greater precision than before.

A Dependency map and Lean-oriented proof sketches

This appendix records the shortest formalization routes for the new paper-level theorems. The snippets are schematic and are not claimed to compile unchanged; theorem names from the repository are shown where known.

A.1 Dependency map

new result	principal existing input	additional mathematics
rational-function rigidity	canonical primitive generator; transcendence of β	injectivity of polynomial evaluation at a transcendental element
product/power criteria	unique (n, k) -coordinates	quadratic coefficient comparison
quotient criterion	unique coordinates	denominator clearing and coefficient comparison
three-smooth output test	transcendence of θ	unique factorization in $\mathbb{N}$
product transcendence	three-smooth-base theorem	classical six exponentials
common power degree	unique coordinates	positive root uniqueness; divisibility
primitive statistics	total lattice count	Möbius inversion
gcd/lcm lattice	exact candidate monoid	prime-valuation min/max
ambient root saturation	primitive valuation gcd of w	Bézout and linear cone arithmetic
optimized differences	arbitrary-order Lean theorem	gamma identity and Stirling

A.2 Multiplication skeleton

A direct statement can use the repository predicate `TwoBaseIntegralSolution`.

```

theorem product_solution_iff_integer_factor
  {x y : R}
  (hx : TwoBaseIntegralSolution x)
  (hy : TwoBaseIntegralSolution y) :
  TwoBaseIntegralSolution (x * y) <->
    x in Set.range ((^): Z -> R) /\
    y in Set.range ((^): Z -> R) := by
  classical
  constructor
  · intro hxy
    by_cases hAE : AlaogluErdosConjecture
    · left
      exact hAE x hx
    · obtain <w, d, a, c, beta, ..., hcoord, ...> :=
        exists_canonical_primitiveGenerator_of_not_alaoğluErdosConjecture hAE
      obtain <nx, kx, hxcoord> := (hcoord x).mp hx
      obtain <ny, ky, hycoord> := (hcoord y).mp hy
      obtain <np, kp, hpcoord> := (hcoord (x * y)).mp hxy
      -- If kx,ky > 0, expansion gives a nonzero quadratic
      -- polynomial over Q vanishing at beta, contradicting
      -- transcendental beta. Therefore kx = 0 or ky = 0.
      ...
  · rintro (hxint | hyint)
    · exact solution_nsmul_left hxint hy
    · exact solution_nsmul_right hyint hx

```

In the actual theorem, the integer range can be simplified to $\mathbb{N}_0$, because every solution is nonnegative. The key algebraic lemma should be phrased independently:

```

lemma transcendental_quadratic_coeff_eq_zero
  {beta a b c : R}
  (hbeta : Transcendental Q beta)
  (h : a * beta^2 + b * beta + c = 0) :
  a = 0 /\ b = 0 /\ c = 0 := ...

```

Mathlib’s polynomial evaluation API may make it cleaner to construct $aX^2 + bX + c$ and apply

the definition of transcendence.

A.3 Output smoothness skeleton

Define a natural predicate

$$\text{ThreeSmooth}(m) :\iff \exists u, v : \mathbb{N}, m = 2^u 3^v.$$

Then prove:

```

theorem integer_iff_outputs_threeSmooth
  {x : R} (hx : TwoBaseIntegralSolution x) :
  x in Set.range ((^) : Z -> R) <->
    ThreeSmooth (natOutputTwo hx) /\
    ThreeSmooth (natOutputThree hx) := by
  constructor
  · rintro <z, rfl>
    ...
  · rintro <<a,b,h2>, <c,d,h3>>
    have hpoly :
      b * logThreeDivLogTwo^2 +
      (a-d) * logThreeDivLogTwo - c = 0 := by
      -- take logarithms of h2 and h3
      ...
    have htheta := transcendental_logThreeDivLogTwo
    have : b = 0 /\ a-d = 0 /\ c = 0 :=
      transcendental_quadratic_coeff_eq_zero htheta hpoly
    ...

```

A practical implementation should avoid partial “natural output” definitions by carrying explicit witnesses $M, A : \mathbb{N}$ for the two integer powers.

A.4 Common-power skeleton

The formal theorem is most reusable if stated for explicit output witnesses.

```

theorem simultaneous_perfectPower_iff_divides_coordinates
  (hfail : not AlaogluErdosConjecture)
  {x beta : R} {n k r : N}
  (hr : 0 < r)
  (hbeta : CanonicalPrimitiveGenerator beta)
  (hxcoord : x = n + k * beta) :
  (Exists fun U : N => (U : R)^r = (2 : R)^x) /\
  (Exists fun V : N => (V : R)^r = (3 : R)^x) <->
    r dvd n /\ r dvd k := by
  -- Convert perfect powers to a solution at x/r,
  -- use unique coordinates, and compare.
  ...

```

The positive-root step can use strict monotonicity of $t \mapsto t^r$ on nonnegative reals or logarithmic injectivity.

A.5 Ambient saturation skeleton

The only nontrivial arithmetic lemma is the primitive valuation reconstruction:

```

lemma primitive_core_root
  {w u q r : N}
  (hw : oddPrimeValuationGCD w = 1)
  (hr : 0 < r)
  (hpow : u^r = w^q) :
  Exists t : N, q = r*t /\ u = w^t := by
  -- Compare every prime factorization coordinate.

```

```
-- Use a Bezout linear combination of the positive valuations of w
-- to prove  $r \mid q$ , then use factorization extensionality.
...
```

The present report’s proof uses $q = dk$. A general lemma of this form would also be useful in `PrimitiveOddCore.lean` and `OutputNormalization.lean`.

A.6 Suggested theorem-order for a pull request

A low-conflict sequence is:

1. add a small `ThreeSmooth` API and output-smoothness theorem;
2. add coordinate comparison lemmas for affine, quadratic, and quotient expressions;
3. prove product, power, and quotient criteria;
4. prove common-power and affine-descent criteria;
5. add candidate divisibility and support results;
6. add primitive-core root reconstruction and ambient saturation;
7. only then add asymptotic and six-exponentials extensions.

This ordering keeps the first several commits elementary and independently reviewable.

B MPFR verifier source

The following is the exact source whose SHA-256 digest is given in [section 11](#).

```
#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <inttypes.h>
#include <math.h>
#include <omp.h>

#if defined(__has_include)
# if __has_include(<mpfr.h>)
#   include <mpfr.h>
#   define HAVE_SYSTEM_MPFR_HEADER 1
# endif
#endif

#ifndef HAVE_SYSTEM_MPFR_HEADER
/* Minimal declarations for the MPFR 4.x ABI. The execution environment had
libmpfr.so.6 but not the development header package. A normal build with
libmpfr-dev installed takes the branch above instead. */
typedef unsigned long mp_limb_t;
typedef long mpfr_prec_t;
typedef long mpfr_exp_t;
typedef int mpfr_sign_t;
typedef struct {
    mpfr_prec_t _mpfr_prec;
    mpfr_sign_t _mpfr_sign;
    mpfr_exp_t _mpfr_exp;
    mp_limb_t *_mpfr_d;
} __mpfr_struct;
typedef __mpfr_struct mpfr_t[1];
typedef __mpfr_struct *mpfr_ptr;
typedef const __mpfr_struct *mpfr_srcptr;
typedef enum {
    MPFR_RNDN = 0, MPFR_RNDZ = 1, MPFR_RNDU = 2,
    MPFR_RNDD = 3, MPFR_RNDA = 4, MPFR_RNDF = 5,
    MPFR_RNDNA = -1
} mpfr_rnd_t;
```

```

extern void mpfr_init2(mpfr_ptr, mpfr_prec_t);
extern void mpfr_clear(mpfr_ptr);
extern int mpfr_set_ui(mpfr_ptr, unsigned long, mpfr_rnd_t);
extern int mpfr_log(mpfr_ptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_mul(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_sub(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_cmp_ui(mpfr_srcptr, unsigned long);
extern double mpfr_get_d(mpfr_srcptr, mpfr_rnd_t);
extern const char *mpfr_get_version(void);
#endif

static inline int is_power_of_two_u64(uint64_t x) {
    return x && ((x & (x - 1)) == 0);
}

typedef struct {
    uint64_t checked;
    uint64_t failures;
    double min_lower;
    double min_upper;
    uint64_t min_lower_m, min_lower_n;
    uint64_t min_upper_m, min_upper_n;
    uint64_t first_fail_m, first_fail_guess;
} Stats;

int main(int argc, char **argv) {
    uint64_t start = 1;
    uint64_t limit = (1ULL << 20);
    int bits = 192;
    if (argc > 1) start = strtoull(argv[1], NULL, 0);
    if (argc > 2) limit = strtoull(argv[2], NULL, 0);
    if (argc > 3) bits = atoi(argv[3]);
    if (start < 1 || limit <= start) {
        fprintf(stderr, "bad range\n");
        return 2;
    }

    int nt = omp_get_max_threads();
    Stats *stats = calloc((size_t)nt, sizeof(Stats));
    if (!stats) return 3;
    for (int i = 0; i < nt; ++i) {
        stats[i].min_lower = INFINITY;
        stats[i].min_upper = INFINITY;
    }

    const long double theta = logl(3.0L) / logl(2.0L);
    double t0 = omp_get_wtime();

    mpfr_t two, three, ln2_lo, ln2_hi, ln3_lo, ln3_hi;
    mpfr_init2(two, bits); mpfr_init2(three, bits);
    mpfr_init2(ln2_lo, bits); mpfr_init2(ln2_hi, bits);
    mpfr_init2(ln3_lo, bits); mpfr_init2(ln3_hi, bits);
    mpfr_set_ui(two, 2, MPFR_RNDN);
    mpfr_set_ui(three, 3, MPFR_RNDN);
    mpfr_log(ln2_lo, two, MPFR_RNDD);
    mpfr_log(ln2_hi, two, MPFR_RNDU);
    mpfr_log(ln3_lo, three, MPFR_RNDD);
    mpfr_log(ln3_hi, three, MPFR_RNDU);

#pragma omp parallel
    {
        int tid = omp_get_thread_num();
        Stats *st = &stats[tid];
        mpfr_t xm, xn, xnp1;
        mpfr_t logm_lo, logm_hi, logn_hi, lognp1_lo;
        mpfr_t lhs_lo, lhs_hi, rhs_lo, rhs_hi, dl, du;
        mpfr_init2(xm, bits); mpfr_init2(xn, bits); mpfr_init2(xnp1, bits);
        mpfr_init2(logm_lo, bits); mpfr_init2(logm_hi, bits);
        mpfr_init2(logn_hi, bits); mpfr_init2(lognp1_lo, bits);
        mpfr_init2(lhs_lo, bits); mpfr_init2(lhs_hi, bits);
        mpfr_init2(rhs_lo, bits); mpfr_init2(rhs_hi, bits);
        mpfr_init2(dl, bits); mpfr_init2(du, bits);
    }

```

```

#pragma omp for schedule(static)
for (uint64_t m = start; m < limit; ++m) {
    if (is_power_of_two_u64(m)) continue;

    /* This is only a locator. The MPFR inequalities below are the proof. */
    uint64_t guess = (uint64_t)floorl(expl(theta * logl((long double)m)));

    mpfr_set_ui(xm, (unsigned long)m, MPFR_RNDN);
    mpfr_log(logm_lo, xm, MPFR_RNDD);
    mpfr_log(logm_hi, xm, MPFR_RNDU);
    mpfr_mul(lhs_lo, logm_lo, ln3_lo, MPFR_RNDD);
    mpfr_mul(lhs_hi, logm_hi, ln3_hi, MPFR_RNDU);

    int ok = 0;
    uint64_t certn = 0;
    double dl_d = 0.0, du_d = 0.0;
    static const int offsets[7] = {0, -1, 1, -2, 2, -3, 3};
    for (int oi = 0; oi < 7; ++oi) {
        int off = offsets[oi];
        if (off < 0 && guess < (uint64_t)(-off) + 1) continue;
        uint64_t n = (uint64_t)((int64_t)guess + off);
        if (n < 1) continue;

        mpfr_set_ui(xn, (unsigned long)n, MPFR_RNDN);
        mpfr_set_ui(xnp1, (unsigned long)(n + 1), MPFR_RNDN);

        /* Directed lower bound for log(m)log(3)-log(n)log(2). */
        mpfr_log(logn_hi, xn, MPFR_RNDU);
        mpfr_mul(rhs_hi, logn_hi, ln2_hi, MPFR_RNDU);
        mpfr_sub(dl, lhs_lo, rhs_hi, MPFR_RNDD);
        if (mpfr_cmp_ui(dl, 0) <= 0) continue;

        /* Directed lower bound for log(n+1)log(2)-log(m)log(3). */
        mpfr_log(lognp1_lo, xnp1, MPFR_RNDD);
        mpfr_mul(rhs_lo, lognp1_lo, ln2_lo, MPFR_RNDD);
        mpfr_sub(du, rhs_lo, lhs_hi, MPFR_RNDD);
        if (mpfr_cmp_ui(du, 0) <= 0) continue;

        ok = 1;
        certn = n;
        dl_d = mpfr_get_d(dl, MPFR_RNDD);
        du_d = mpfr_get_d(du, MPFR_RNDD);
        break;
    }

    if (!ok) {
        ++st->failures;
        if (!st->first_fail_m || m < st->first_fail_m) {
            st->first_fail_m = m;
            st->first_fail_guess = guess;
        }
        continue;
    }

    ++st->checked;
    if (dl_d < st->min_lower) {
        st->min_lower = dl_d;
        st->min_lower_m = m;
        st->min_lower_n = certn;
    }
    if (du_d < st->min_upper) {
        st->min_upper = du_d;
        st->min_upper_m = m;
        st->min_upper_n = certn;
    }
}

mpfr_clear(xm); mpfr_clear(xn); mpfr_clear(xnp1);
mpfr_clear(logm_lo); mpfr_clear(logm_hi);
mpfr_clear(logn_hi); mpfr_clear(lognp1_lo);
mpfr_clear(lhs_lo); mpfr_clear(lhs_hi);
mpfr_clear(rhs_lo); mpfr_clear(rhs_hi);
mpfr_clear(dl); mpfr_clear(du);

```

```

}

Stats total = {0};
total.min_lower = INFINITY;
total.min_upper = INFINITY;
for (int i = 0; i < nt; ++i) {
    Stats *s = &stats[i];
    total.checked += s->checked;
    total.failures += s->failures;
    if (s->min_lower < total.min_lower) {
        total.min_lower = s->min_lower;
        total.min_lower_m = s->min_lower_m;
        total.min_lower_n = s->min_lower_n;
    }
    if (s->min_upper < total.min_upper) {
        total.min_upper = s->min_upper;
        total.min_upper_m = s->min_upper_m;
        total.min_upper_n = s->min_upper_n;
    }
    if (s->first_fail_m &&
        (!total.first_fail_m || s->first_fail_m < total.first_fail_m)) {
        total.first_fail_m = s->first_fail_m;
        total.first_fail_guess = s->first_fail_guess;
    }
}

double elapsed = omp_get_wtime() - t0;
printf("mpfr_version=%s\n", mpfr_get_version());
printf("range=[%" PRIu64 ",%" PRIu64 "]\n", start, limit);
printf("precision_bits=%d\nthreads=%d\n", bits, nt);
printf("checked_nonpowers=%" PRIu64 "\n", total.checked);
printf("failures=%" PRIu64 "\n", total.failures);
printf("min_lower_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
        total.min_lower, total.min_lower_m, total.min_lower_n);
printf("min_upper_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
        total.min_upper, total.min_upper_m, total.min_upper_n);
if (total.failures) {
    printf("first_failure=(m,guess)=(%" PRIu64 ",%" PRIu64 ")\n",
            total.first_fail_m, total.first_fail_guess);
}
printf("elapsed_seconds=%.6f\n", elapsed);

mpfr_clear(two); mpfr_clear(three);
mpfr_clear(ln2_lo); mpfr_clear(ln2_hi);
mpfr_clear(ln3_lo); mpfr_clear(ln3_hi);
free(stats);
return total.failures ? 1 : 0;
}

```

A standard build on a system with MPFR development headers is

```

gcc -O3 -fopenmp mpfr_scan.c -lmpfr -lm -o mpfr_scan
OMP_NUM_THREADS=5 ./mpfr_scan 1 67108864 192

```

The recorded audit used sixteen shorter invocations to produce independently inspectable chunk logs.

C Complete chunk audit

The canonical chunk log has SHA-256 digest

```
dcaa7f3b7e5be44acaee9015a2045da14ee42d8fa2d96b9619cdd44c072a6fd7
```

and contains the full stdout of every invocation. The table reports the exact scanned integer interval, number of non-powers certified, and the smallest directed lower and upper logarithmic margins in each chunk. Every failure count was zero.

Table 1: Directed-rounding audit through 2^{26} .

chunk	integer range	checked	min. lower gap	min. upper gap
0	1–4,194,303	4,194,281	1.1539×10^{-17}	3.0076×10^{-17}
1	4,194,304–8,388,607	4,194,303	4.6319×10^{-20}	1.3515×10^{-19}
2	8,388,608–12,582,911	4,194,303	1.2811×10^{-18}	1.3515×10^{-19}
3	12,582,912–16,777,215	4,194,304	4.6319×10^{-20}	2.9082×10^{-19}
4	16,777,216–20,971,519	4,194,303	1.1898×10^{-18}	1.3515×10^{-19}
5	20,971,520–25,165,823	4,194,304	5.4212×10^{-19}	1.5111×10^{-19}
6	25,165,824–29,360,127	4,194,304	4.6319×10^{-20}	1.7669×10^{-19}
7	29,360,128–33,554,431	4,194,304	1.4439×10^{-19}	5.4132×10^{-21}
8	33,554,432–37,748,735	4,194,303	4.2839×10^{-19}	1.3515×10^{-19}
9	37,748,736–41,943,039	4,194,304	3.9292×10^{-19}	1.0460×10^{-19}
10	41,943,040–46,137,343	4,194,304	1.9177×10^{-19}	1.5111×10^{-19}
11	46,137,344–50,331,647	4,194,304	1.0117×10^{-19}	3.4145×10^{-19}
12	50,331,648–54,525,951	4,194,304	5.6927×10^{-20}	1.2322×10^{-19}
13	54,525,952–58,720,255	4,194,304	3.6447×10^{-20}	1.1345×10^{-19}
14	58,720,256–62,914,559	4,194,304	6.7779×10^{-20}	5.4132×10^{-21}
15	62,914,560–67,108,863	4,194,304	5.0923×10^{-20}	3.9197×10^{-20}
total		67,108,837		

The global lower-gap record occurred at (58,570,438, 2,048,703,434,093); the global upper-gap record occurred at (29,411,704, 687,581,893,443). The repeated upper gap in chunk 14 comes from exact scaling by two on the input and by three on its θ -power.

D Reproducibility checklist

To reproduce the finite claim independently:

1. obtain MPFR 4.2.x and GMP on a 64-bit platform;
2. save the source in [section B](#) and verify its SHA-256 digest;
3. compile with optimization and OpenMP, or remove OpenMP pragmas for a serial run;
4. run the sixteen half-open intervals listed in [table 1](#) at 192 bits;
5. verify that every invocation reports `failures=0`;
6. sum `checked_nonpowers` and obtain 67,108,837;
7. compare the two global extremal records;
8. optionally rerun at 256 or 320 bits and on a second MPFR build;
9. for a theorem-grade result, translate the intervals into exact rational certificates and replay them in Lean rather than trusting the executable.

The finite statement is deterministic. Thread count changes execution time and reduction order for the recorded minimum metadata, but not whether an individual directed inequality is certified.

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